Information Retrieval Project Report:
Data Collection from Jumia Website

1. Introduction

This report outlines the project undertaken to collect and analyze data from the Jumia website, focusing specifically on television products. The data collected includes essential details such as product name, price, customer ratings, category, sales rank, product brand, and images. This information is valuable for understanding market trends, consumer preferences, and competitive analysis in the e-commerce sector.

2. Project Objectives

The primary objectives of this project are:

Data Collection: Extract comprehensive and accurate data on televisions listed on the Jumia website.

Data Analysis: Analyze the collected data to identify patterns and insights regarding pricing, customer preferences, and market dynamics.

Tool Utilization: Demonstrate the effective use of web scraping tools, specifically BeautifulSoup, for data extraction.

Market Insights: Provide actionable insights that can help stakeholders make informed decisions.

3. Planned Approach for Data Scraping and Analysis

3.1 Data Collection

3.1.1 Target Website

The target website for this project is Jumia, a leading e-commerce platform in Africa. The focus is on the television product category due to its high consumer interest and diverse range of offerings.

3.1.2 Tools and Technologies

BeautifulSoup: A Python library used for parsing HTML and XML documents. It provides Pythonic idioms for iterating, searching, and modifying the parse tree.

3.1.3 Data Fields:

The following data fields were identified as critical for the analysis:

Product Name: The name of the television.

Price/Price Range: The listed price or price range of the television.

Customer Ratings/Reviews: Ratings and reviews provided by customers.

Category: The category under which the television is listed.

Sales Rank: The rank of the television in terms of sales on the platform.

Product Brand: The brand of the television.

Product Image/Images: URLs of the product images.

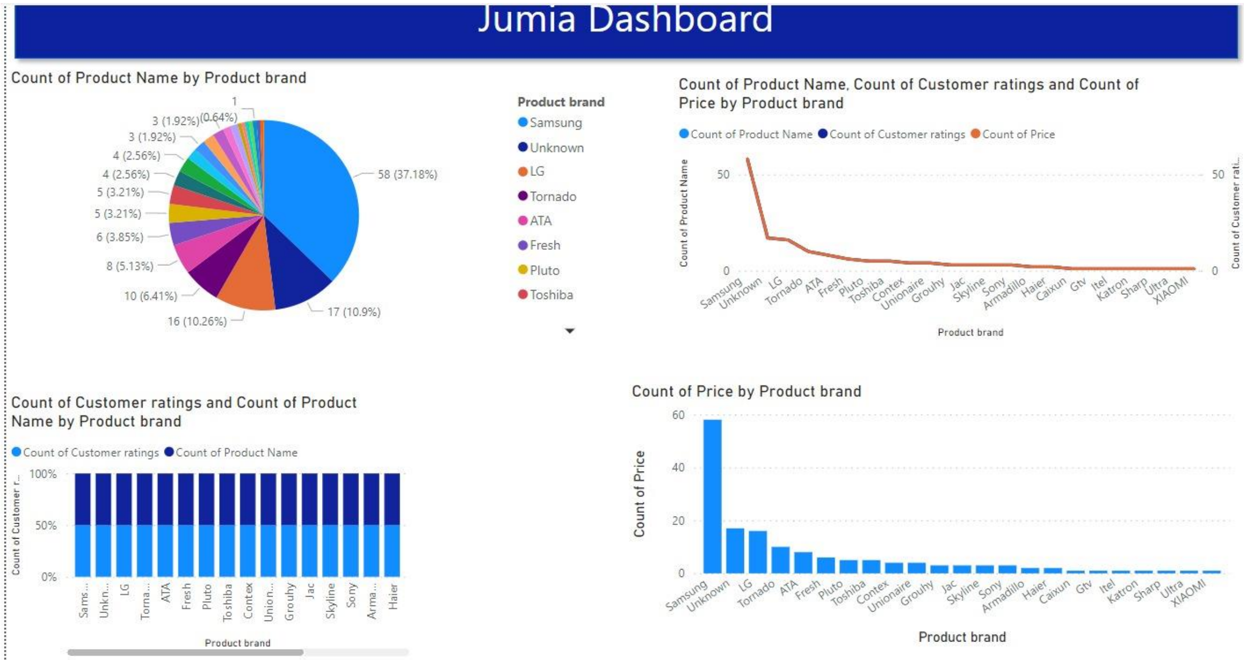
3.1.4 Data Extraction Process

Sending Requests: Use the requests library to send HTTP GET requests to the Jumia website and retrieve the HTML content.

Parsing HTML: Utilize BeautifulSoup to parse the retrieved HTML and navigate the DOM tree to locate the relevant data fields.

Data Extraction: Extract the required data fields using BeautifulSoup's methods such as find, find_all, and CSS selectors.

3.2 Data Analysis



3.2.1 Data Cleaning and Preprocessing

Handling Missing Values: Identify and address missing or incomplete data.

3.2.2 Exploratory Data Analysis (EDA)

Descriptive Statistics: Compute summary statistics for numerical fields such as price and ratings.

Data Visualization: Create visualizations (e.g., bar charts, histograms) to identify trends and patterns.

Category Analysis: Analyze the distribution of products across different categories and brands.

3.2.3 Advanced Analysis

Price Comparison: Compare prices across different brands and models to identify pricing strategies.

Sentiment Analysis: Analyze customer reviews to gauge overall sentiment and identify common themes or issues.

Sales Rank Correlation: Examine the relationship between sales rank, price, and customer ratings.

4. Expected Outcomes

The project aims to deliver a comprehensive dataset and insightful analysis of televisions on the Jumia website. The outcomes will include:

A cleaned and structured dataset ready for further analysis or machine learning applications.

Visual and statistical analysis revealing market trends and consumer preferences.

Actionable insights for stakeholders, such as identifying popular brands, optimal pricing strategies, and common customer concerns.

5. Conclusion

This project leverages web scraping techniques to gather valuable data from Jumia, demonstrating the practical application of BeautifulSoup for data extraction. The analysis of this data will provide meaningful insights into the television market, contributing to informed decision-making for businesses and consumers alike.